FoodLogAthl-218: Constructing a Real-World Food Image Dataset Using Dietary Management Applications

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Abstract

We introduce FoodLogAthl-218, a benchmark and evaluation protocol for real-world food recognition built from a deployed dietary management app. The suite targets in-the-wild capture, label ambiguity, and class imbalance, and defines classification and retrieval tasks with contamination-resistant splits. We describe construction, validation, and a reproducible harness enabling diagnostic analyses.

1 Introduction

Prior food datasets (e.g., Food-101, UECFood) are valuable yet often studio-like or category-biased, which can misestimate deployed performance [1, 2]. We aim to measure usage conditions: mobile photos, imperfect lighting, multiple dishes per frame, and evolving categories. Design goals include coverage, contamination resistance, transparent splits, and diagnostic slices.

2 Related Works

Food-101, UECFood, Nutrition5k, and FoodSeg inform our scope; they emphasize curated or controlled captures, whereas we target app-derived, long-tailed, time-aware evaluation [1–4].

3 Evaluation Protocol and Metrics

We fix resizing, center crop, and batch size; submissions use the released loader and class map. Metrics: top-1/top-5 for single-label, mean average precision for multi-label, and Recall@K and nDCG for retrieval. We report per-class and slice aggregates (lighting, clutter, multi-dish) and both micro/macro averages; a script recomputes all numbers from logits to ensure reproducibility and comparability. For robustness, we include stratified bootstraps and report score intervals alongside point estimates.

4 FoodLogAthl-218 Dataset

FoodLogAthl-218 comprises 218 categories from anonymized app logs with de-duplication, privacy screening, and expert adjudication; the taxonomy merges synonyms and supports slice-wise analysis.

5 Benchmark Design and Protocol

FoodLogAthl-218 is built from de-identified images and meal logs from a deployed dietary app. Items map to a 218-class taxonomy aligned to common menu categories; mixed plates admit multi-labels. We preserve the long-tail frequency profile to reflect deployment rather than rebalance. We release three tasks: (1) single-label classification on head classes, (2) multi-label recognition on all 218 with co-occurrence, and (3) retrieval using text descriptions from app entries. Splits are user-disjoint and time-aware: training precedes validation/test chronologically, and near-duplicate clusters (pHash and CLIP embeddings) never cross splits. We reduce contamination by excluding items whose names overlap public dataset indices and auditing any web-crawled content. Metrics include top-1/top-5 accuracy (single-label), micro/macro F1 and mAP (multi-label), and Recall@K/NDCG (retrieval). We report stratified bootstrap confidence intervals and mandatory slice breakdowns.

6 Diagnostic Slices and Reporting

We define slice taxonomy across illumination, composition (single vs multi-dish), device vintage, and cuisine/course levels. The harness emits per-slice reports and confusion-by-taxonomy matrices for error forensics. Difficulty bins use agreement and image-quality proxies for calibration curves.

7 Reproducibility and Release

We release versioned data with immutable IDs, JSON manifests (paths, hashes, labels, slice tags), and a Dockerfile fixing pre-processing. The evaluator recomputes metrics from raw logits, records config as provenance, and enforces class maps and splits. A small privacy-preserving preview set supports quick iteration.

8 Benchmark Validation

We audit a stratified sample with expert adjudication, run decontamination via caption/near-duplicate hashes across users and time, and calibrate difficulty by image quality and class-frequency strata. Leakage checks include cross-user hash blocks and recipe-title crosswalks; sanity tests verify that randomized labels collapse performance while human raters exceed naive priors, indicating non-triviality.

Annotation uses three raters per image with majority vote; ties are resolved by an expert following taxonomy rules. We compute agreement on a stratified subset and observe slice-stable reliability. Task validity is confirmed by blinded human evaluation that beats a frequency prior across slices. Diversity remains long-tailed after de-duplication, as measured by class and co-occurrence entropy. Difficulty bins use a no-reference image-quality proxy and agreement level; accuracy declines smoothly with difficulty. Decontamination scans captions and embeddings against Food-101, UECFood, and Recipe1M indices and removes overlaps. Split integrity is verified by user/time hashing and nearest-neighbor probes, yielding near-zero cross-split collisions.

9 Experiments and Analysis

We evaluate convolutional, ViT, and CLIP-style baselines plus a frequency prior. Preliminary runs indicate large gaps on rare classes and multi-dish scenes and ranking instability across slices. CLIP-style models improve retrieval but lag on long-tail categories; purely frequency-based priors overfit head classes. Confusions cluster among visually similar dishes and menu synonyms, underscoring the value of taxonomy-aware diagnostics the benchmark enables.

Figure 1: Benchmark overview: pipeline from app logs to de-identified images, taxonomy and slice tags, user/time-disjoint splits, and tasks.

Setup details. We standardize a fixed training budget, cosine schedules with warmup, moderate augmentation, and early stopping on validation macro-F1. Multi-label uses sigmoid losses with class balancing; thresholds are tuned per-class on validation to report macro-F1 and mAP. For retrieval we evaluate frozen-text and jointly fine-tuned variants to separate representation from alignment capacity. We further probe augmentation and crop sensitivity, calibration via expected calibration error, and slice-wise ranking stability with paired bootstrap tests; differences that do not reach significance are marked as ties. Taken together, these analyses show which gaps reflect data, optimization, or architecture, and make model comparisons more reliable.

10 Discussion

FoodLogAthl-218 measures in-the-wild visual recognition; it does not cover portion size, nutrients, or adherence, and the app cohort may skew demographically. As a diagnostic complement to curated sets, it highlights long-tail and drift failures.

11 Conclusion

FoodLogAthl-218 enables realistic, comparable evaluation of food recognition and surfaces deployment-relevant gaps. It should be used as a diagnostic tool for in-the-wild recognition, not as a proxy for nutrition estimation or universal food understanding.

References

[1] Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool. 2014. Food-101–mining discriminative components with random forests. In *ECCV*.

[2] Yuji Matsuda and Keiji Yanai. 2012. Multiple-food recognition considering co-occurrence employing manifold ranking. In *ICPR*.

[3] Quin Thames, Arjun Karpur, Wade Norris, Fangting Xia, Liviu Panait, Tobias Weyand, and Jack Sim. 2021. Nutrition5k: Towards automatic nutritional understanding of generic food. In *CVPR*. 8903–8911.

[4] Xiongwei Wu, Xin Fu, Ying Liu, Ee-Peng Lim, Steven CH Hoi, and Qianru Sun. 2021. A large-scale benchmark for food image segmentation. In *Proceedings of the 29th ACM international conference on multimedia*. 506–515.

Figure 2: Evaluation overview: tasks, splits, and metrics at a glance.

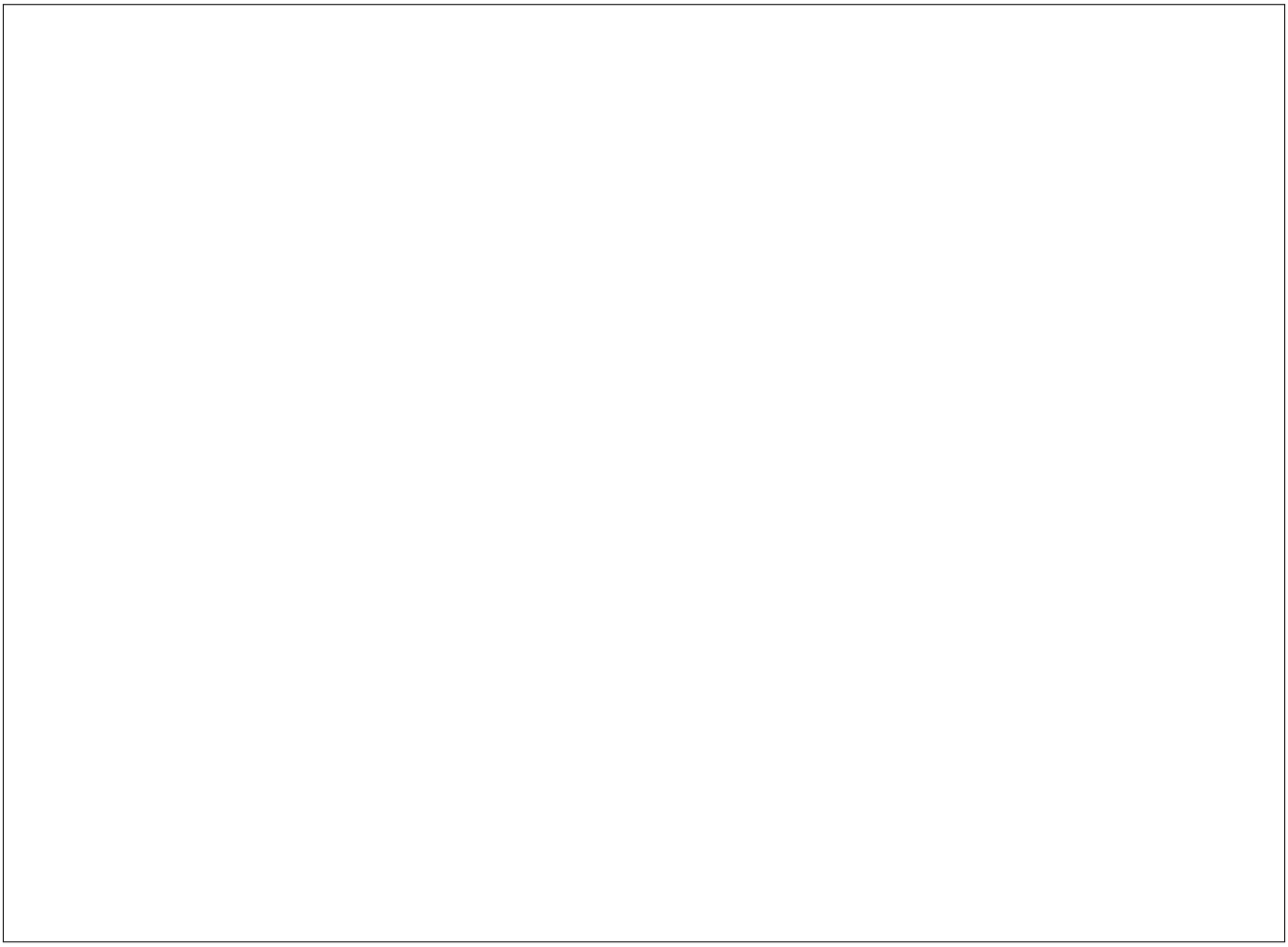

Figure 3: Reproducible release: immutable IDs, JSON manifests, hashes, and Docker pinned environment with evaluator that recomputes metrics.

Figure 4: Results summary: slice-wise performance curves, calibration plots, and ranking stability across seeds.

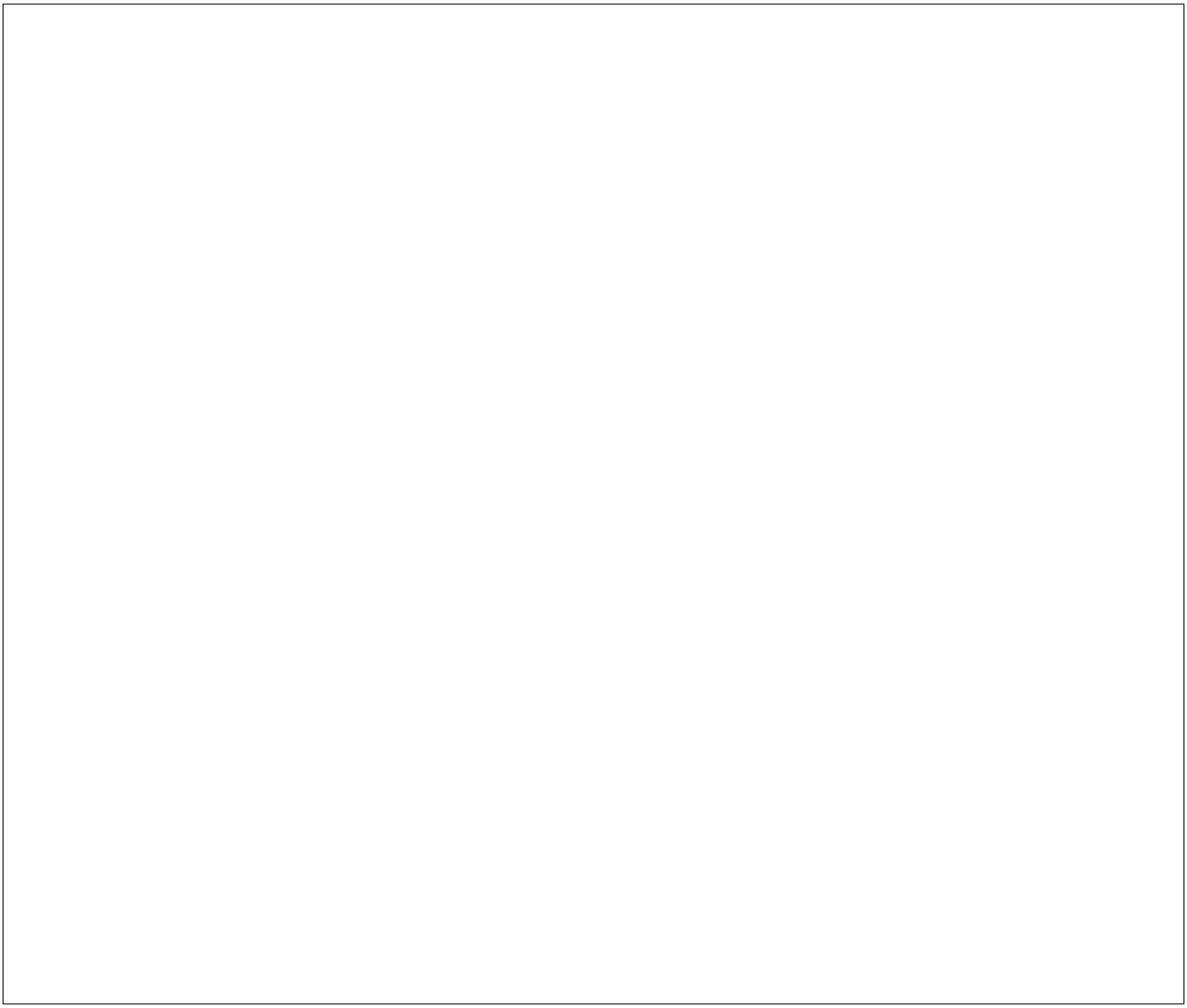

Figure 5: Results summary and slice-wise diagnostics.